India

GROWTH ACCOUNTING

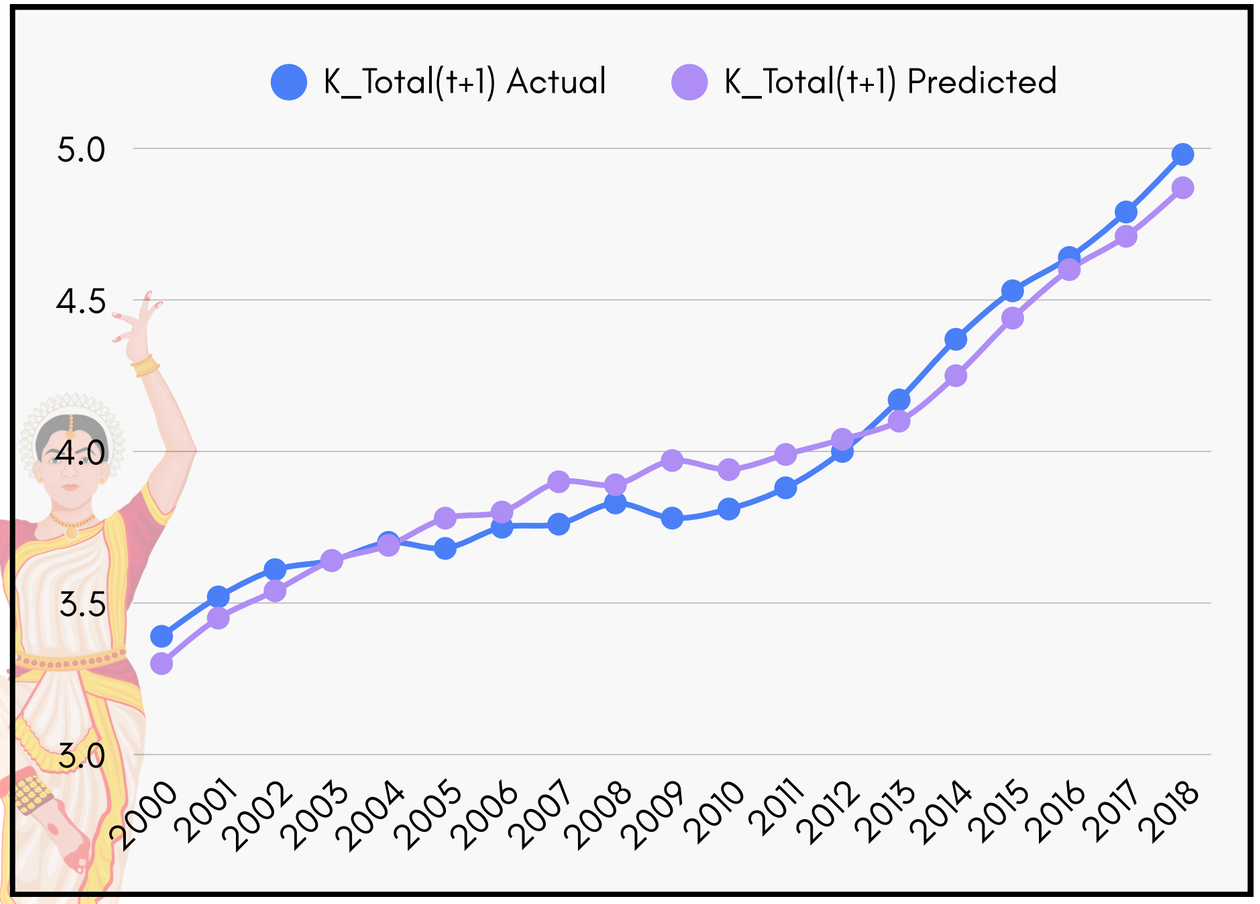


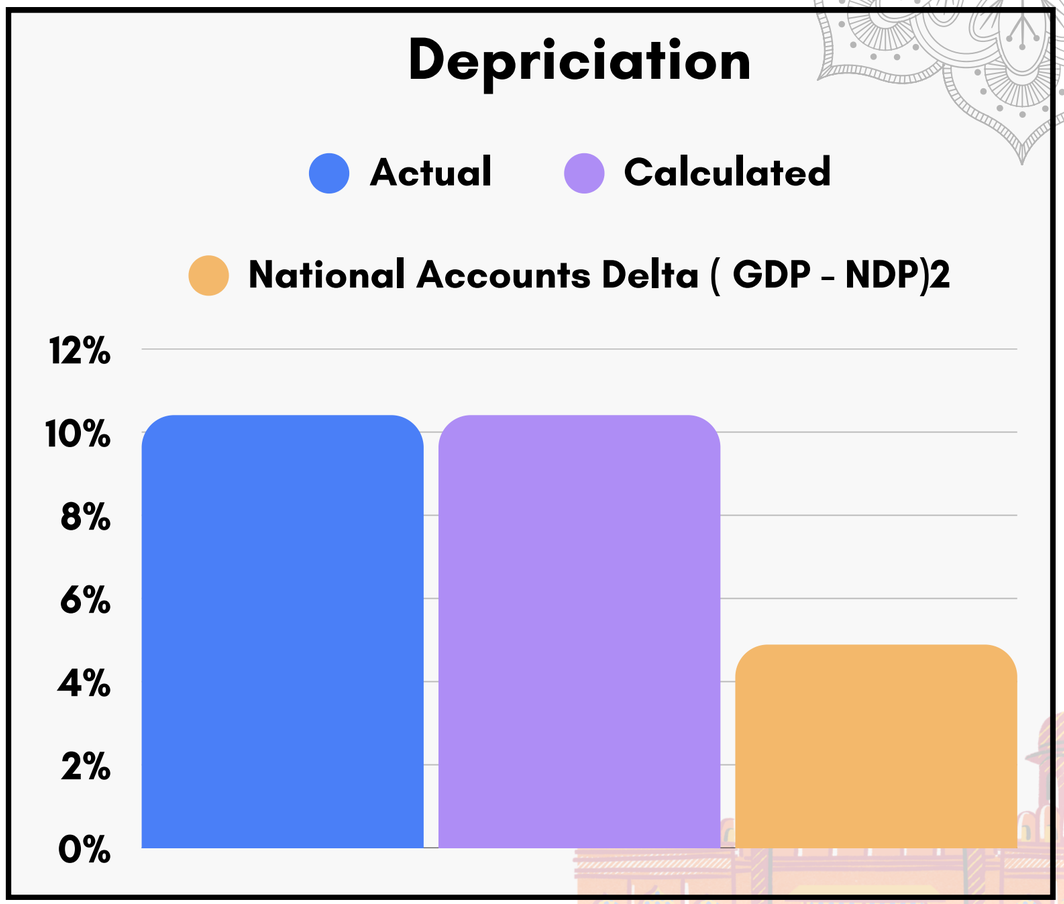
Verifying the the capital accumulation equation.

$$K_{t+1}=(1-\delta)K_t+I_t$$

Method of Getting Delta (δ)	Value of Delta (δ)	Inference
Regression	**10.4072% or (0.10472)**	India's capital stock wears out at 10.41% per year as directly estimated from 19 years of actual investment and capital data — statistically definitive with near-zero probability of being wrong.
World Bank Data	**10.4279% or (0.104079)**	The independently published World Bank rate of 10.43% matches the regression to within 0.02 percentage points — confirming both methods are measuring the same true depreciation phenomenon.
GDP-NDP (Average of all Years)	**4.89% or (0.489)**	The national accounts rate of 4.89% is less than half the other two because India's large informal economy has assets counted in the capital stock denominator but their depreciation is never recorded in the numerator - making India's capital appear to last twice as long as it actually does.

Actual Vs Calculated (Capital Stock)





Verifying the the Production Function

$$Y=A.K^{\alpha}.L^{\beta}$$

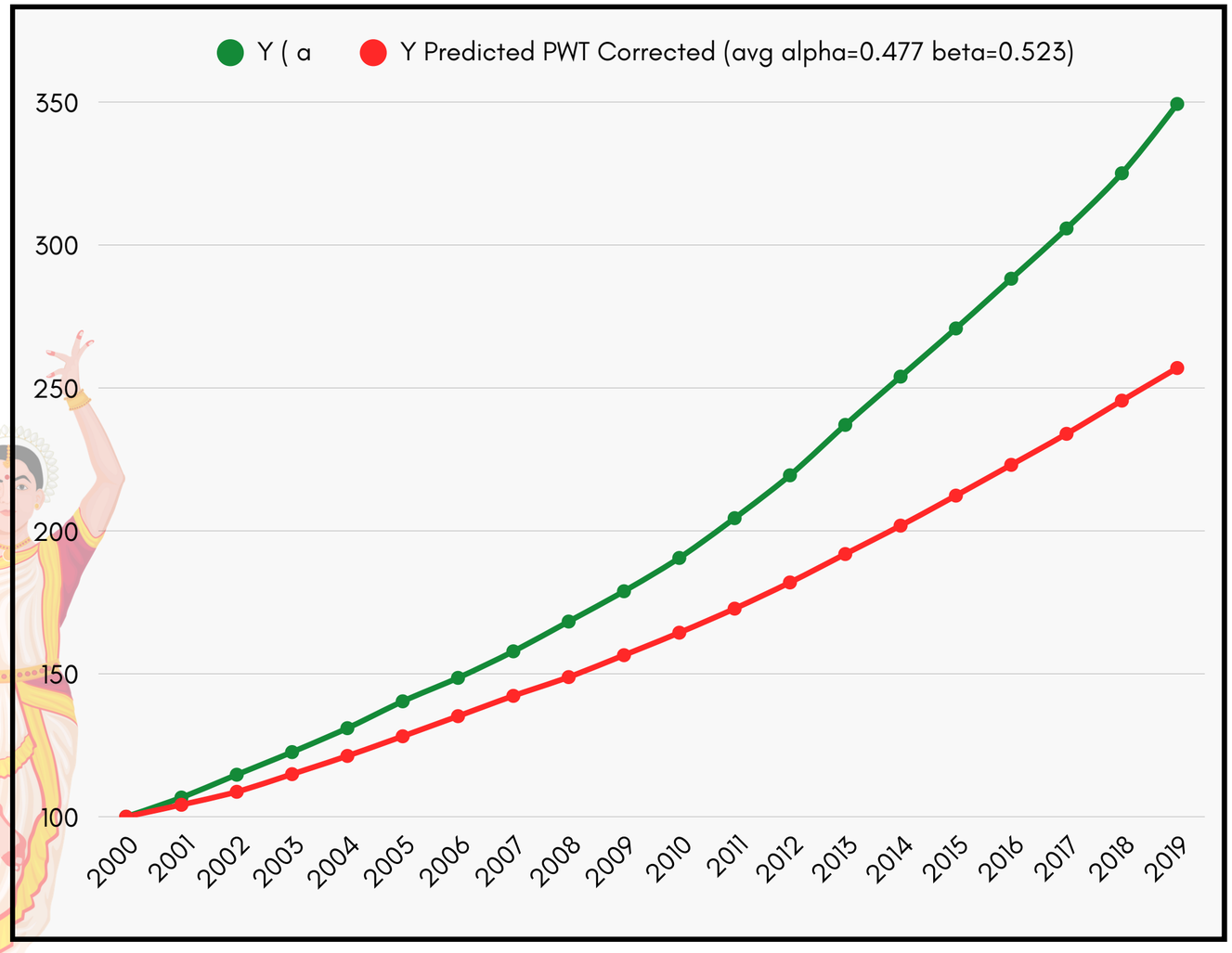
$$\log Y = \log A + \alpha \log K + \beta \log L$$

Method of Getting Alpha	Alpha (α)	Beta (β)	Alpha + Beta	Inference
Regression (Assuming CRS)	**0.640591**	**1.199418**	**1.840009**	Capital and labour in India did not just grow in quantity — they grew in quality and connectivity. Every new highway made existing roads more valuable. Every new factory worker trained other workers around them. These network and spillover effects mean the whole is greater than the sum of its parts, and OLS captures this as a sum of elasticities above 1.
Regression (Individually Regression)	**0.791449**	**0.208551**	**1**	When you constrain the sum to 1, all the increasing returns that Row 1 was picking up get forced into alpha. Capital absorbs all the scale effects, network effects and embodied technology that Row 1 spread across both coefficients. So alpha looks inflated compared to the PWT value.
PWT STAT (Average)	**0.477046**	**0.522954**	**1**	PWT measures what capital actually earns, not how much it contributes to output at the margin. Informal sector workers in India earn very little — their labour share in national income is small. But they contribute substantially to output. The regression picks up their output contribution but PWT only counts their income share, so the two measures diverge.

Verifying the Production Function

$$Y=A.K^{\alpha}.L^{\beta}$$

$$\log Y = \log A + \alpha \log K + \beta \log L$$



- IRS ($\alpha+\beta=1.84$) amplifies factor growth more than PWT sourced ($\alpha+\beta=1.00$) — higher combined elasticity means each unit of K and L growth produces more output under IRS
- Both start at 100 in 2000 and diverge every single year — the gap is not a one-time event but a compounding structural feature of India's economy
- PWT sourced reflects income shares — what capital and labour actually earned. IRS reflects marginal productivity — how much output each factor actually generated. The two measure different things
- The 92 point gap by 2019 represents India's unmeasured scale advantage — infrastructure network effects, urbanisation agglomeration and technology embodied in capital imports that income share data cannot capture
- Neither line is wrong — IRS describes India's actual growth dynamics, PWT sourced describes the neoclassical benchmark. India clearly outperformed the neoclassical prediction throughout 2000-2019

Verifying the the Production Function

$$Y=A.K^{\alpha}.L^{\beta}$$

$$\log Y = \log A + \alpha \log K + \beta \log L$$

Methods	A(t)	Inferences
Using α used from Regression	**0.9790**	The average TFP index of **0.9790** (base year 2000 = 1) indicates that total factor productivity in India declined marginally over 2000–2018, standing about **2.1% below its 2000 level** on average. This is derived using the regression-estimated capital share **α = 0.7914** (from the Cobb-Douglas per-worker specification). The high α reflects capital deepening as the dominant driver of India's output growth, leaving limited residual to TFP. The result implies that India's GDP growth over this period was largely **input-driven** - explained by capital and labour accumulation - rather than by efficiency or technological progress. TFP actually contracted during 2008–2013, consistent with the global financial crisis and domestic policy headwinds, and only partially recovered post-2014.
Using α used from ILO Stat	**0.9145**	The average TFP index of 0.9145 using year-specific PWT-sourced factor shares ($\alpha \approx 0.477$–0.534, $\beta \approx 0.466$–0.523) shows a steeper average decline of roughly 8.6% below the 2000 baseline. Assigning a larger weight to labour (higher β) means that the slower growth of labour relative to output is penalised more heavily in the TFP residual, resulting in a lower measured A(t). This corroborates the finding that India's productivity performance was weak — the economy absorbed more capital and labour without a commensurate rise in efficiency. The persistent downward trend through 2013 followed by a modest recovery post-2014 suggests structural reforms and digitalisation may have begun to lift TFP, but not enough to restore the 2000 benchmark within the sample period.